34 When a beam of positively charged alpha particles is directed towards a thin sheet of gold in a vacuum, the following observations are made.

1 Most of the alpha particles pass straight through the sheet of gold.

2 A small number of alpha particles are deflected through angles greater than 4°.

Which row links each observation to its appropriate conclusion?

	observation 1	observation 2
A	The atom consists mainly of empty space.	The negative charge is concentrated at the centre of the atom.
B	The atom consists mainly of empty space.	The positive charge is concentrated at the centre of the atom.
C	The negative charge is concentrated at the centre of the atom.	The atom consists mainly of empty space.
D	The positive charge is concentrated at the centre of the atom.	The atom consists mainly of empty space.